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Euclidean Geometry In Mathematical Olympiads

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Problem 1.46 (Canada 1997/4)


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illogical_21

1034 posts

Mar 17, 2019, 1:41 AM • 2 👍

#1

The point O is situated inside the parallelogram $ABCD$ such that $\angle AOB + \angle COD = 180^\circ$. Prove that $\angle OBC = \angle ODC$.

illogical_21

1034 posts

Mar 17, 2019, 1:41 AM • 3 👍

#2

Solution

Let O' be a point such that $ADOO'$, $BCOO'$ are parallelograms, then $\angle BOA = \angle BO'A \implies AOBO'$ cyclic, so $\angle OBC = \angle BOO' = \angle BAO' = \angle CDO$, so we are done.

fjm30

474 posts

Sep 9, 2019, 8:08 AM • 1 👍

#3

illogical_21 wrote:

Solution

What is the motivation behind constructing the point O'

This post has been edited 1 time. Last edited by fjm30, Sep 9, 2019, 8:08 AM

kootrapali

4527 posts

Mar 1, 2020, 4:38 PM

#4

Solution

Let P be the point on AB such that $OP \parallel BC$. Let O' and P' be the images of O and P , respectively, after a translation of $\triangle AOB$ mapping A to D and B to C . We have O', P', O, P collinear, and since $O'P' = OP$, then $O'O = P'P = BC$. Also, $OB = O'C$, so $O'OBC$ is a parallelogram. Now we see that $\angle DOC + \angle DO'C = \angle DOC + \angle AOB = 180^\circ$, so $DOCO'$ is cyclic. Finally, $\angle OBC = \angle OO'C = \angle ODC$.

DPS

812 posts

Mar 2, 2020, 5:37 AM

#5

fjm30 wrote:

illogical_21 wrote:

Solution

What is the motivation behind constructing the point O'

To create a cyclic quadrilateral. It is a well known construction.

Williamgolly

3760 posts

Apr 26, 2020, 1:53 PM

#6

Hi yall! I used directed angles in my proof but the book didnt. Why not? Arent there multiple configurations?

DPS

Apr 26, 2020, 2:36 PM • 1 👍

#7

812 posts



Williamgolly wrote:

Hi yall! I used directed angles in my proof but the book didnt. Why not? Arent there multiple configurations?

Hi Williamgolly! Since it's given that O is inside the $\triangle ABC$, so only one configuration is possible.

Mathsciencecl...

1241 posts

Apr 26, 2020, 5:57 PM • 3 👍

It's still okay to use directed angles though 😊. Good practice.

#8

Jerry122805

1074 posts

Aug 13, 2020, 10:49 PM

Solution**Solution with directed angles**

#9